

Vegetation index-based crop coefficients to estimate evapotranspiration by remote sensing in agricultural and natural ecosystems

Edward P. Glenn,^{1*} Christopher M. U. Neale,² Doug J. Hunsaker³ and Pamela L. Nagler⁴

¹ Environmental Research Laboratory of the University of Arizona, 2601 East Airport Drive, Tucson, AZ, 86706, USA

² Civil & Environmental Engineering, Utah State University, Logan, UT, USA, 84322-4110

³ USDA-ARS, Arid Land Agricultural Research Center, 21881 N. Cardon Lane, Maricopa, AZ, 85138, USA

⁴ U.S. Geological Survey, Sonoran Desert Research Station, University of Arizona, Tucson, AZ, USA, 85721

Abstract:

Crop coefficients were developed to determine crop water needs based on the evapotranspiration (ET) of a reference crop under a given set of meteorological conditions. Starting in the 1980s, crop coefficients developed through lysimeter studies or set by expert opinion began to be supplemented by remotely sensed vegetation indices (VI) that measured the actual status of the crop on a field-by-field basis. VIs measure the density of green foliage based on the reflectance of visible and near infrared (NIR) light from the canopy, and are highly correlated with plant physiological processes that depend on light absorption by a canopy such as ET and photosynthesis. Reflectance-based crop coefficients have now been developed for numerous individual crops, including corn, wheat, alfalfa, cotton, potato, sugar beet, vegetables, grapes and orchard crops. Other research has shown that VIs can be used to predict ET over fields of mixed crops, allowing them to be used to monitor ET over entire irrigation districts. VI-based crop coefficients can help reduce agricultural water use by matching irrigation rates to the actual water needs of a crop as it grows instead of to a modeled crop growing under optimal conditions. Recently, the concept has been applied to natural ecosystems at the local, regional and continental scales of measurement, using time-series satellite data from the MODIS sensors on the Terra satellite. VIs or other visible-NIR band algorithms are combined with meteorological data to predict ET in numerous biome types, from deserts, to arctic tundra, to tropical rainforests. These methods often closely match ET measured on the ground at the global FluxNet array of eddy covariance moisture and carbon flux towers. The primary advantage of VI methods for estimating ET is that transpiration is closely related to radiation absorbed by the plant canopy, which is closely related to VIs. The primary disadvantage is that they cannot capture stress effects or soil evaporation. Copyright © 2011 John Wiley & Sons, Ltd.

KEY WORDS MODIS; Landsat; reflectance-based crop coefficients; irrigation scheduling; hydrological cycle; climate change studies

Received 18 July 2011; Accepted 17 October 2011

INTRODUCTION

Starting in the 1960s, operational methods for estimating evapotranspiration (ET) from land surfaces began to diverge (Shuttleworth, 2007). Meteorological approaches emphasized modeling land-atmosphere boundary conditions, whereas the agricultural community developed a more empirical approach based on crop coefficients (K_c) (Jensen, 1968; Jensen *et al.*, 1971; Allen *et al.*, 1998, 2005). A crop coefficient relates the actual ET of a crop at a given stage of development to reference ET (ET_o) calculated from meteorological data by one of a number of different equations, depending upon data availability. If sufficient weather data are available, as from automated meteorological stations, the FAO Penman–Monteith formula for daily ET_o can be used (Allen *et al.*, 1998):

$$ET_o = \frac{[0.408\Delta(R_n - G) + \gamma 900 / (T + 273) u_2 (e_s - e_a)]}{[\Delta + \gamma(1 + .34u_2)]} \quad (1)$$

where:

ET_o reference evapotranspiration [mm day^{-1}],
 R_n net radiation at the crop surface [$\text{MJ m}^{-2} \text{day}^{-1}$],
 G soil heat flux density [$\text{MJ m}^{-2} \text{day}^{-1}$],
 T mean daily air temperature at 2 m height [$^{\circ}\text{C}$],
 u_2 wind speed at 2 m height [m s^{-1}],
 e_s saturation vapour pressure [kPa],
 e_a actual vapour pressure [kPa],
 $e_s - e_a$ saturation vapour pressure deficit [kPa],
 D slope vapour pressure curve [$\text{kPa } ^{\circ}\text{C}^{-1}$],
 g psychrometric constant [$\text{kPa } ^{\circ}\text{C}^{-1}$].

Equation (1) is derived from the general form of the Penman–Monteith equation, as presented in the FAO-56 manual on crop evapotranspiration (Allen *et al.*, 1998). Numerical constants for surface and aerodynamic resistances and albedo are inserted into the formula based on a

*Correspondence to: Edward P. Glenn, Environmental Research Laboratory of the University of Arizona, 2601 East Airport Drive, Tucson, AZ 86706 USA.
E-mail: eglenn@ag.arizona.edu

hypothetical grass crop with an assumed height of 0.12 m, an albedo of 0.23 and a fixed bulk canopy resistance of 70 s m^{-1} (Allen *et al.*, 1998). Numerous other formulas for calculating ET_o exist (Xu and Singh, 2002; Fisher *et al.*, 2011). Alfalfa is used as the reference crop in various areas of the United States (Wright, 1982; Allen *et al.*, 2005), which is typically 1–1.25 times higher than ET_o as defined in FAO-56 (Allen *et al.*, 1998). The American Society of Civil Engineers modified Equation (1) to include different crop heights (Walter *et al.*, 2002).

Once ET_o is computed, actual ET (simply abbreviated as ET in this paper) is then calculated as:

$$ET = K_c ET_o \quad (2)$$

Crop coefficients are generally developed through field experiments using lysimeters (e.g. Jensen, 1968), planted with the crop being studied and are typically at monthly intervals to conform to the main growth stages of the crop. These crop coefficients are usually developed for crops grown under optimum agronomic conditions, and are therefore only useful approximations of the actual ET and water requirements for a given crop. However, the actual crop ET in the field can vary from estimated K_c -based ET for a number of reasons, including crop variety differences, planting density, climatic factors, nutrient status, irrigation management, salinity and other conditions. Often, these factors reduce actual water demand below that expected for a crop grown under optimal conditions. Therefore, using crop coefficients for irrigation scheduling can lead to over-irrigation of crops, which can be a serious concern at the field level when irrigation water is in short supply, as well as at the district and regional level of water management, especially in water-short arid and semi-arid areas of the world (Santos *et al.*, 2007).

Vegetation Index (VI) methods replace (or supplement) crop coefficients with a VI that reflects the actual growth stage of the crop at the time of measurement. VI-based crop coefficients (K_{c-VI}) have been developed for individual and mixed crops in agricultural regions starting nearly 30 years ago (e.g. Neale, 1983). More recently, the concept has been applied to natural ecosystems at local, regional and global scales of measurement (Glenn *et al.*, 2010).

The following sections review the different VIs commonly used in ET studies; the theoretical justification for K_{c-VI} methods; their application to individual and mixed crops in agriculture; and their extension to natural ecosystems. A final section reviews the sources of error and uncertainty inherent in these measurements. The overall goal is to describe the methods in sufficient detail that they can be adapted to other applications by an increasing number of resource managers and researchers who need accurate estimates of ET at spatial and temporal scales that require remote sensing methods. This review builds on other recent reviews on the use of remote sensing to estimate ET, which include other approaches such as the use of thermal bands and methods based on solving the surface energy balance equation (Diak *et al.*,

2004; Courault *et al.*, 2005; Neale *et al.*, 2005; Allen *et al.*, 2007; Glenn *et al.*, 2007; Gowda *et al.*, 2008; Kalma *et al.*, 2008; Kustas and Anderson, 2009; Tang *et al.*, 2009; Glenn *et al.*, 2010).

VEGETATION INDICES USED IN ET STUDIES

VIs were developed with the launch of the first satellites used for vegetation monitoring in the early 1970s (Rouse *et al.*, 1973; reviewed in Bausch and Neale, 1987; Bannari *et al.*, 1995; Huete and Glenn, 2011). The Normalized Difference VI (NDVI) is the most common VI, and it is calculated as:

$$NDVI = (\rho_{NIR} - \rho_{Red}) / (\rho_{NIR} + \rho_{Red}) \quad (3)$$

where ρ_{NIR} and ρ_{Red} are the near infrared and red band reflectance values received by a sensor over a vegetated surface, respectively. NDVI values fall between -1.0 and $+1.0$, with water having negative values, soil slightly positive values, and vegetation having increasingly high values approaching 0.95 for very dense vegetation cover (Bannari *et al.*, 1995). The NDVI captures the sharp contrast in reflection of light from green leaves between the Red and NIR wavelengths. Red light is strongly absorbed by chlorophyll a and b in leaves, interacting mostly with the top layers of a dense canopy, but nearly all of the NIR is transmitted, reflected or scattered by the mesophyll structure in leaves, penetrating deep into the canopy and interacting with multiple leaf layers. This light is received by a downward-looking sensor sensitive to the NIR band. Numerous studies have shown a high correlation between NDVI and biophysical characteristics of plants, including fractional cover (f_c), leaf area index (LAI); chlorophyll content; and wet and dry biomass (reviewed in Bausch and Neale, 1987; Glenn *et al.*, 2008). NDVI is also strongly correlated with physiological processes that depend on light absorption by the canopy, including ET, yield, and net and gross primary productivity.

Two other VIs used in the ET studies reported here are the Soil Adjusted VI (SAVI) and the Enhanced VI (EVI) (Nagler *et al.*, 2001; Huete *et al.*, 2002; Huete and Glenn, 2011). SAVI is calculated as:

$$SAVI = (\rho_{NIR} - \rho_{Red})(1 + L) / (\rho_{NIR} + \rho_{Red} + L) \quad (4)$$

where L is a correction factor usually set at 0.5 to account for soil background effects. EVI is calculated as:

$$EVI = G(\rho_{NIR} - \rho_{Red}) / (\rho_{NIR} + C_1 \times \rho_{Red} + C_2 \times \rho_{Blue} + L) \quad (5)$$

where C_1 and C_2 are coefficients designed to correct for aerosol resistance, which uses the blue band to correct for aerosol influences in the red band. The coefficients, C_1 and C_2 , are set at -6 and 7.5 , respectively, G is a gain factor (set at 2.5) and L for this model is a canopy background adjustment (set to 1.0). These VIs are described in detail in (Huete *et al.*, 2002; Huete and Glenn, 2011). EVI and SAVI are non-linearly related to

NDVI and extend the range over which VIs respond to increases in foliage density beyond a LAI of about 3 (Choudhury *et al.*, 1994; Nagler *et al.*, 2001; Huete *et al.*, 2002).

The MODIS LAI product (Myneni *et al.*, 2002) uses visible and NIR data collected at several view angles and tabular data on land cover type to calculate LAI based on a radiation transfer model. There is a strong but non-linear relation between MODIS LAI and NDVI and other VIs.

EXPERIMENTAL AND THEORETICAL JUSTIFICATION FOR K_{C-VI} METHODS

In early work on this topic, Neale (1983), Bausch and Neale (1987), Neale (1987), and Bausch and Neale (1989), Neale *et al.*, (1989) and Bausch (1995) compared crop development characteristics with ET and NDVI values for irrigated corn crops at two experimental farm sites in Colorado. Crops were grown in fields equipped with weighing lysimeters, and K_c was calculated with alfalfa as the reference crop grown at each facility (i.e. $K_c = ET_{\text{corn}} / ET_{\text{alfalfa}}$). NDVI was measured with down-looking radiometers suspended 2 m above the canopies. They found that NDVI was highly correlated with LAI and f_c , and that K_{C-VI} derived from radiometric measurements closely tracked measured K_c over the crop cycle (Figure 1). The inflection point for maximum ET, maximum LAI and maximum NDVI occurred at approximately the same points in the growing cycle (ca. DOY 210–220). They concluded (Bausch and Neale, 1987) that crop coefficients derived from NDVI were independent of time parameters and other factors affecting crop performance, and represented real-time coefficients that could be used to schedule irrigations on a field-by-field basis.

Choudhury *et al.* (1994) used a modeling approach to justify using VIs to replace K_c in Equation (2). They showed that:

$$T_c = (VI^*)^\eta \quad (6)$$

where T_c is a plant transpiration coefficient (by analogy a basal crop coefficient, K_{cb}) and VI^* is a VI stretched between 0 (representing bare soil) and 1.0 (representing fully transpiring, unstressed vegetation) calculated using the formula:

$$VI^* = 1 - (VI_{\text{max}} - VI) / (VI_{\text{max}} - VI_{\text{min}}) \quad (7)$$

where VI_{max} is the VI value when ET is maximal and VI_{min} is the VI of bare soil. Then, ET (ignoring bare soil evaporation) is approximated as:

$$ET = ET_o (VI^*)^\eta \quad (8)$$

The exponent, η , is determined by the relationship between transpiration and the VI used in Equation (7). Choudhury *et al.* (1994) pointed out that SAVI (and by extension EVI from which it is derived) extends the range of LAI values over which a VI is responsive, because it

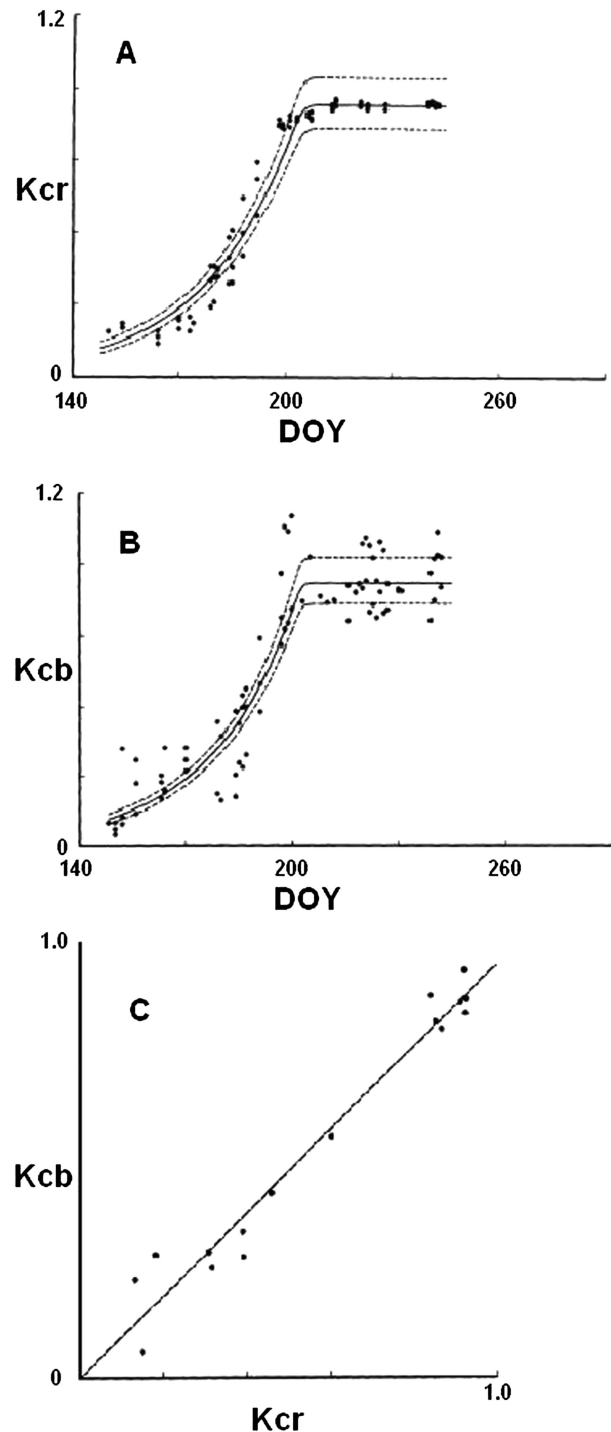


Figure 1. NDVI-determined crop coefficient (K_{cr}) (A); lysimeter-determined crop coefficient (K_{cb}) (B); and regression of lysimeter-determined (y-axis) and NDVI-determined crop coefficients (C) for corn crops grown at Fruita, Colorado, 1984 (from Neale *et al.*, 1989)

does not 'saturate' as quickly as NDVI at high LAIs. They showed that SAVI was curvilinearly related to LAI up to LAI of 5, and since ET is also curvilinearly related to LAI, ET was near-linearly related to SAVI. Hence, for EVI and SAVI, the exponent η in Equation (8) is 1.0. By contrast, NDVI, which typically saturates at an LAI of about 3.0, was curvilinearly related to ET, and η was less than 1.0. SAVI also had a higher r^2 relationship to measured ET than NDVI.

Another advantage of using the SAVI or EVI for crop coefficient relationships is that these indices are less sensitive to soil background reflectance changes due to surface moisture, a common occurrence in irrigated areas. Darkened, wet soils will artificially increase the NDVI during the vegetative growth stages when incomplete canopy cover is prevalent, leading to an over prediction of ET when using a relationship developed with dry soil surface data. This was observed by Neale (1987) in the development of NDVI – basal crop coefficient (K_{cb}) relationship and led to the use of the SAVI by Bausch (1993) for corn, Neale *et al.* (1996) for cotton; Jayanthi *et al.* (2000) and Jayanthi *et al.* (2007) for beans and potato, respectively. The basal crop coefficient, proposed by Wright (1982) for several crops in southern Idaho, is defined for a dry soil surface and no water limitation in the root zone of the crop. Thus, the SAVI may be better suited to be related to the K_{cb} with the added advantage that relationships developed with the SAVI using an $L=0.5$ can be applied to different regions with dark or light natural dry soil reflectance (Huete, 1988). However, a definitive experiment directly comparing the different VIs with respect to f_c , LAI or ET has not yet been conducted, to our knowledge.

APPLICATION TO INDIVIDUAL CROPS

Following the early work on corn (Neale, 1983, 1987; Bausch and Neale, 1987; Neale *et al.*, 1989), the reflectance-based crop coefficient methods (K_{c-VI} methods) have been applied to numerous other crops, as illustrated by the following examples.

Vineyards and orchards

Vineyards and orchards are especially difficult to model with static crop coefficients due to differences in plant spacing, varieties and plant heights among fields, and the variable amount of bare soil or understory vegetation between vines or trees. Furthermore, ET changes during fruit filling. Campos *et al.* (2011) assessed satellite-based K_{c-VI} s for irrigated grapes in Spain, using an energy-balance moisture flux tower to independently measure actual ET and ET_o . Both NDVI and SAVI were able to predict ET by simple linear regression equations, validating the use of remote sensing to estimate ET in wine grapes.

Samani *et al.* (2009) used NDVI from Landsat imagery to develop field-scale K_{c-VI} values for pecan orchards in the lower Rio Grande Valley, New Mexico, USA. Pecans cover 11,000 ha and consume 40% of irrigation water in the area, and there is a need to quantify actual water use by the trees to resolve issues of water rights allocation, water management and agronomic practices. Satellite-derived ET values were within 4% of ET measured at an eddy covariance tower site in a mature field. They found that ET over 279 fields varied from 498 to 1259 mm year⁻¹, and that only 5% of fields were within the range of maximum ET and K_c set for the crop by expert

opinion, indicating the potential for significant water savings. Although many of the fields were under chronic water stress, ET was still closely related to NDVI because the trees adjusted LAI to reflect water availability.

Wheat

Wheat has been extensively studied due to its importance as an irrigated crop in water-short arid and semi-arid regions El-Raki *et al.* (2007). Duchemin *et al.* (2006) used NDVI from two Landsat images to map LAI and transpiration in irrigated wheat fields in Morocco. The satellite-derived K_{c-VI} had an error of only 15% compared to ground measurements and the maps displayed a high field-to-field variability in actual water requirements depending on the state of canopy development, opening up the possibility of improving irrigation efficiency through satellite monitoring of crop phenology. Gontia and Tiwari (2010) used NDVI and SAVI from satellite sensors to map wheat water use in an irrigation district in West Bengal, India. They developed field-specific crop coefficients based on measured ET_o and satellite data, and were able to generate spatially distributed, monthly ET maps of the irrigation district. These maps could then be compared to water application rates to identify areas where water savings were possible.

Extensive work has been conducted on wheat by the USDA-ARS in a desert irrigation district in Maricopa, Arizona, USA (Hunsaker *et al.*, 2005b; Hunsaker *et al.*, 2007a, 2007b). In initial field experiments (Hunsaker *et al.*, 2005b), a crop coefficient model for wheat ET was developed based on the dual crop coefficient approach of FAO-56, which includes a basal crop coefficient, K_{cb} , corresponding to crop transpiration, and a separate coefficient, K_e , that accounts for soil evaporation (Allen *et al.*, 2005):

$$ET = K_{cb}ET_o + K_eET_o \quad (9)$$

The experimental results showed that the contribution of soil evaporation was approximately 10% of the total wheat ET in this irrigation district, as surface soils were normally dry and precipitation was low. The NDVI-derived K_{c-VI} tracked within 5% of measured K_{cb} (Figure 2). In subsequent experiments, wheat was grown in field plots for two seasons under six treatment combinations: typical, dense and sparse plant densities, with adequate and limited nitrogen. The FAO-56 reference-crop method was used to calculate K_{cb} and ET in Equation (9) on daily time steps to predict soil moisture depletion and to schedule irrigations. For comparison, K_{cb} was estimated by the NDVI-derived K_{c-VI} derived from the earlier experiments (Hunsaker *et al.*, 2007a). In these studies, NDVI was measured with a hand-held radiometer two to four times per week in all treatment plots. Actual soil moisture depletion was measured frequently with neutron hydroprobes in access tubes ports installed in plots.

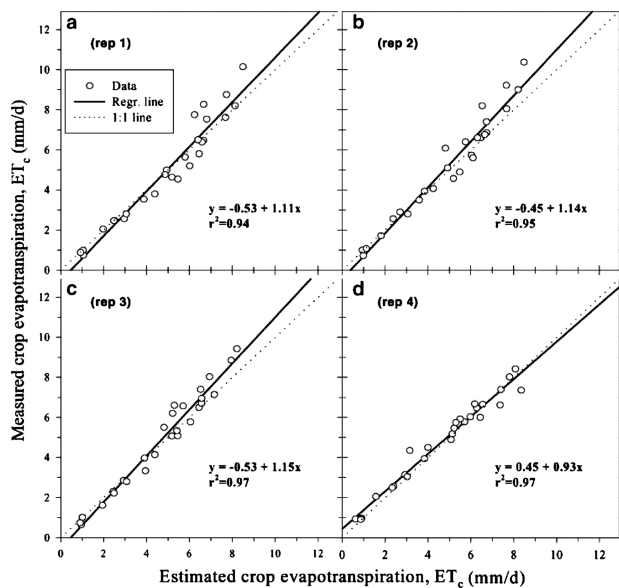


Figure 2. Regression of measured ET to ET estimated by NDVI-derived crop coefficients for wheat grown in lysimeters at Maricopa, AZ (from Hunsaker *et al.*, 2005)

The NDVI method gave more accurate predictions of actual irrigation demands than the FAO-56 method under all treatment conditions ($P < 0.05$) (Hunsaker *et al.* 2007a, 2007b). For dense stands of wheat receiving adequate fertilization, the FAO-56 method tended to underestimate irrigation requirements by about 10%, whereas for the sparsely planted, nitrogen-limited plots, the FAO-56 method overestimated irrigation requirements by up to 52%. The NDVI method had a root mean square error (RMSE) of about 15% of measured water use with no bias towards under or overestimation across treatment. Over both years and all treatments, water use efficiency based on grain yield and water applications was 10% higher for the NDVI method than for the FAO-56 method ($P < 0.05$), with a bigger difference for sparsely planted, nitrogen-limited crops. Hence, under actual field conditions, in which crops are frequently growing under suboptimal conditions, use of NDVI measurements as a supplement to static crop coefficients can result in significantly improved irrigation efficiency.

Cotton

Cotton is a high-water-use crop grown extensively in arid irrigation districts around the world. As a row crop, it is difficult to model with conventional crop coefficients due to variations in plant spacing, row orientation and plant density among fields. Hunsaker *et al.* (2003, 2005a) developed NDVI-based basal crop coefficients for cotton grown under three levels of plant density and two levels of nitrogen management typical of variations in agronomic practices in Maricopa, Arizona. The conventional FAO-56 K_{cb} curve performed reasonably well for sparse crops but consistently underestimated crop water demand for the higher plant densities, whereas the K_{c-VI} method gave consistently accurate results across the range of

treatments. They concluded that the NDVI approach has the potential to improve the accuracy of irrigation scheduling when weather, plant density or other factors cause cotton canopy development and water use patterns to depart from typical conditions.

The above studies were conducted with aerial photography and ground-based radiometers. Hunsaker *et al.* (2009) subsequently reduced the number of aerial images needed to monitor cotton water demand using a sampling protocol in which a few aerial acquisitions made at key points in the crop cycle in conjunction with 20 or less ground measurements of NDVI could provide a cost-effective method for estimating the spatial distribution of cotton water use on a field-by-field basis, improving irrigation and water scheduling efficiency.

Gonzalez-Dugo and Mateos (2008) used ground radiometers and Landsat NDVI to model ET in cotton fields in southern Spain. Ground and satellite observations of canopy reflectance gave similar results, with SAVI preferred over NDVI due to its linear relationship with ET. Over a large sampling of fields, ET was proportional to biomass, but lint yields were substantially lower than predicted by a linear model of biomass to yield. Evapotranspired water productivity for lint ranged from 0.3 to 0.78 kg m⁻³, indicating that considerable water savings might be realized by careful irrigation scheduling based on NDVI-derived crop models.

Other crops

Using hand-held radiometers, high-resolution digital aerial imagery, or satellite imagery, NDVI-based crop coefficients have also been developed for potato (Jayanthi *et al.*, 2007, broccoli (El-Shikha *et al.*, 2007), sugarbeet (Jayanthi, 2003; Gonzalez-Dugo and Mateos, 2008), soybean (Gonzalez-Dugo *et al.*, 2009), the oilseed crop camelina (Hunsaker *et al.*, 2011), sorghum (Singh and Irmak, 2009) and alfalfa (Singh and Irmak, 2009). All these studies reported positive results. The NDVI-derived K_{c-VI} for alfalfa differed from the FAO-56 K_c in southern Nebraska (Singh and Irmak, 2009), but no ground measurements of ET were available for comparison so it was uncertain whether the error was in K_{c-VI} or K_c calculated by the FAO-56 procedure.

WIDE AREA ESTIMATES OVER MIXED CROPS

The need to develop crop-specific coefficients is a serious drawback for using remote sensing for ET monitoring over whole agricultural districts. As a general rule, satellite imagery cannot be used to map specific crops without extensive field observation and careful timing of the imagery, so crop maps need to be constructed and validated using a combination of ground observations, farmer interviews and remote sensing methods. However, Choudhury *et al.* (1994) pointed out that the ET:VI relationship was not necessarily crop specific, and that Equation (2) might be used over mixed crops without a

serious loss of accuracy. Allen and Pereira (2009) found a reasonable agreement between measured K_c and f_c over a wide range of tree crops, and the relationship was improved by including plant height in the regression. The ground ET data set were limited in that study. However, similar findings were reported by Trout *et al.* (2008) for tree and vegetable crops, and a close correlation was noted between Landsat-derived f_c based on NDVI, and f_c measured on the ground over 30 fields with crops ranging from trees (almonds, pistachios), to vines (grapes), and row crops (onions, tomatoes, cantaloupes, watermelon, beans, pepper, garlic and lettuce (Figure 3). The only aberrant crop was red lettuce, which has a low NDVI due to reflection of red light from anthocyanin in the leaf cell vacuoles.

Rafn *et al.* (2008) compared an NDVI method for deriving K_c to the thermal-band METRIC method over mixed crops in an Idaho irrigation district. They concluded that the NDVI/ K_c method was a fully objective and repeatable process that was fast, easy and less costly to employ than the METRIC method. They projected that the method could be applied outside the area where it was developed. Similarly, Gonzalez-Dugo *et al.* (2009) found that NDVI-derived crop coefficients combined with ET_0 measurements predicted ET of corn and soybean crops as well as thermal-band methods in Central Iowa, although it overestimated ET for corn during a dry-down period. These findings open up the possibility of monitoring ET without field-by-field knowledge of cropping patterns.

Murray *et al.* (2009) compared an NDVI approach to estimating ET with a K_c approach for irrigation districts on the Lower Colorado River. The K_c approach used four annual Landsat images and ground methods to map cropping patterns and derive acreages for each crop in the irrigation districts, then used K_c values set by expert opinion (Jensen, 1998) to calculate monthly values of ET. Murray *et al.* (2009) replaced the individual K_c with a K_{c-VI} based on a regression analysis between ground

measurements of ET and a MODIS-EVI-derived K_{c-VI} for alfalfa and several species of riparian plants found in the irrigation districts. The modeled K_{c-VI} closely matched the K_c expected for alfalfa. When applied over five irrigation districts along the river, the K_{c-VI} and K_c methods gave similar mean estimates of ET, but values differed among the irrigation districts. Different agronomic practices in the districts produced different patterns of ET that were not captured by static crop coefficients, whereas K_{c-VI} detected differences in crop phenology that presumably led to differences in ET. MODIS provides 16-day composite images based on near-daily overpasses, capturing the details of crop development needed for irrigation scheduling, compared to less frequent imagery available from Landsat. The 250-m resolution of MODIS NDVI and EVI products is sufficient for monitoring the relatively large fields in the Lower Colorado River irrigation districts, but would be inadequate for fields less than about 25 ha, which dominate many agricultural districts.

An NDVI-based crop monitoring program is also in development for California's irrigation districts as part of the Terrestrial Observation and Prediction System (Johnson, 2010). The goal of the program is to provide daily ET products with a 1-day lag, with spatial resolution sufficient to be used for irrigation scheduling of individual fields. The methodology is based on the strong correlation that has been noted between Landsat-derived NDVI and f_c for several crops and maturity levels (Trout *et al.*, 2008). Landsat images are acquired from the EROS Data Center in near real time and processed to give atmospherically corrected surface reflectances for each band. Then, NDVI values are calculated and converted on a per-pixel basis to f_c and K_{cb} using linear relationships developed by Trout *et al.* (2008) for Central Valley crops. The output maps and data are available to growers through a variety of Web interfaces. The system has been successfully tested for irrigation scheduling for vineyards in the San Joaquin Valley and is now being extended to cover all of the state's irrigation districts.

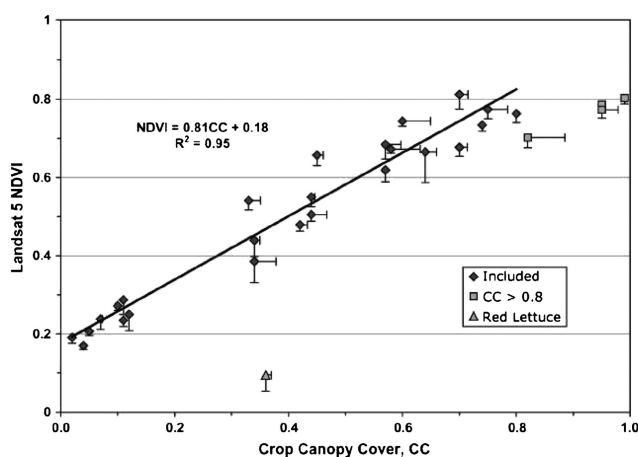


Figure 3. Relationship between Landsat 5 NDVI and multispectral camera canopy cover (CC) and the linear regression line for data represented by diamonds. Bars are standard deviations. Squares represent high CC crops (> 0.8), and the triangle represents a field of red lettuce, with low NDVI due to reflection of red light from anthocyanins in vacuoles (from Trout *et al.*, 2008)

K_{c-VI} METHODS FOR NATURAL ECOSYSTEMS

Obstacles to applying K_{c-VI} methods to natural ecosystems

K_{c-VI} methods assume that there is a fixed or at least predictable relationship between ET_0 and leaf-level transpiration. The VI term can then be used to project leaf-level transpiration rates to ground-area estimates by the relationships between VI, f_c and LAI, which in many cases are linear and might not be crop specific. Implicit in the method is the assumption that stomatal conductance (G_s) can be treated as a constant in daily time steps and that plants are not subject to stresses that differentially reduce G_s on a spatial or temporal basis over the area of interest. While this is often true for irrigated crops, it might not be true for plants in natural ecosystems. Mata-Gonzalez *et al.* (2005) objected to the application of crop coefficients to natural ecosystems and thought they could

overestimate plant water use by 50%–100% in arid regions where heat and moisture stress are common.

On the other hand, optimization theory predicts that plants will adjust LAI to match the environment's capacity to support photosynthesis (Field, 1991; Field *et al.*, 1995; Albrizio and Steduto, 2003). Seasonal leaf production in an ecosystem is generally matched to the limiting factors for photosynthesis at that location (e.g. nutrients, water and radiation). Even in sparse deserts, rainfall tends to be efficiently captured by vegetation, and across agricultural and natural ecosystem, transpiration usually accounts for 70% to > 90% of ET (Huxman *et al.*, 2004; Williams *et al.*, 2004). As a result, measured ET rates (as from energy balance flux towers) are usually strongly correlated with satellite-derived VIs over a wide range of biomes. Therefore, estimating f_c or LAI by remote sensing is a good first step in deriving ET estimates over mixed vegetation types in natural ecosystems (Glenn *et al.*, 2010).

The example of saltcedar as a test of assumptions

Saltcedar (*Tamarix ramosissima*) is an introduced, salt-tolerant shrub that has spread widely on western U.S. rivers. Because it grows in dense stands, it has been considered to have very high water use, and a number of studies have attempted to measure saltcedar ET. Based on LAI, Jensen (1998) set a crop coefficient for saltcedar equal to alfalfa on the Lower Colorado River. However, Westenburg *et al.* (2006), using Bowen ratio moisture flux towers in dense stands at Havasu National Wildlife Refuge on the Lower Colorado River, reported rates of about 800 mm year⁻¹, less than half of ET_o or alfalfa. Neale *et al.* (2001) reported midday rates equal to less than half of net radiation for a dense stand on the Middle Rio Grande as measured by moisture flux towers, and the measured rates were only half as great as rates estimated by aerial remote sensing methods.

Groeneveld *et al.* (2007) and Groeneveld and Baugh (2007) estimated saltcedar ET from NDVI based on a simple crop coefficient method, in which NDVI was stretched between bare soil (0) and fully transpiring crop fields (1.0) on Landsat images. The scaled NDVI was multiplied by ET_o to estimate ET on an annual basis from single summer Landsat scenes, and the results were closely related to values measured at moisture flux towers ($r^2 = 0.95$). Since they have access to a constant source of water, phreatophytes are considered to be non-stressed with relatively constant rates of G_s on daily time steps at a given location (e.g. Groeneveld and Baugh, 2007; Groeneveld *et al.*, 2007).

However, Nagler *et al.* (2009a, 2009b) found a wide range of values for G_s , leaf-level transpiration (E_L) and ground-level transpiration (E_G) among six saltcedar stands on the Lower Colorado River at Blythe (Figure 4). The stands were scattered over a river terrace and were given fanciful names based on summer working conditions. Saltcedar at Slitherin (Figure 4A) had a diurnal E_L response that tracked the radiation curve, indicating

relatively unstressed conditions (Xu and Shen, 2005). However, plants at the other five sites showed moderate to severe midday depression of both E_L and G_s . Furthermore, plants at all sites had significant rates of nighttime transpiration, as noted in other studies with saltcedar (Moore *et al.*, 2008).

Figure 5 shows plant-specific LAI (LAPS), f_c , E_L and E_G at the six sites. E_L and E_G (Figure 5C,D) were very low at the Hot Springs site due to the intrusion of geothermal water into the root zone. Excluding this stand, E_L ranged from 1.7 to 2.8 mm m⁻² leaf d⁻¹ among sites with a coefficient of variation of about 20% around the mean value. Factors that controlled E_L at the sites were salinity, depth to the water table and soil texture, which determined rates of hydraulic conductivity from the capillary fringe into the plant roots.

Nagler *et al.* (2009a, 2009b) regressed ET/ET_o against MODIS-derived EVI for a variety of riparian species as well as alfalfa on the Lower Colorado River and obtained a reasonable overall relationship that could be used for estimating ET of crops and riparian vegetation with a standard error of about 20% of the mean value (Figure 6). They concluded that a remote sensing estimate of ET based on the mean value of E_L could be reasonably accurate for the biome as a whole, but might not accurately predict E_G of any specific stand or agricultural field due to variations in G_s . This conclusion is probably generally true for remote sensing estimates of ET by K_c -VI methods in natural ecosystems.

Examples of other ET-VI algorithms in natural ecosystems

Despite the caveats stated above, quite a number of successful VI methods have been developed to estimate ET in a wide variety of landscape types, including rainforests (Juarez *et al.*, 2008), arctic tundra and boreal forests (Zhang *et al.*, 2009b), riparian zones (Nagler *et al.*, 2005a, 2005b; 2009a,b; Murray *et al.*, 2009; Scott *et al.*, 2008; Barz *et al.*, 2009), desert phreatophyte communities (Groeneveld and Baugh, 2007; Groeneveld *et al.*, 2007), semi-arid grasslands (Nagler *et al.*, 2007; Alfieri *et al.*, 2009) and shrublands (Nagler *et al.*, 2007), sparse desert shrublands (Glenn *et al.*, 2008), and mixed landscapes at the regional (Wang *et al.*, 2007; Juarez *et al.*, 2008; Leuning *et al.*, 2008; Wang and Liang, 2008; Boegh *et al.*, 2009; Li *et al.*, 2009; Scheffeld *et al.*, 2009; Zhang *et al.*, 2009a), continental (Zhang *et al.*, 2006; Cleugh *et al.*, 2007; Guerschman *et al.*, 2009) and global (Mu *et al.*, 2011; Fisher *et al.*, 2008; Jung *et al.*, 2010) scales of measurement. Examples are in Table I.

The examples in Table I vary in complexity but have some features in common. First, nearly all of them use the MODIS sensors on the Terra satellite, and EVI has been preferred over NDVI. The Terra satellite provides near-daily coverage of the earth, and MODIS pixels have a resolution of 250 m in the Red and NIR and 500 m in the Blue, an improvement over the AVHRR system with 1-km resolution (Huete *et al.*, 2002). Furthermore, data are available as georeferenced and atmospherically corrected VI or LAI products, in which composites of

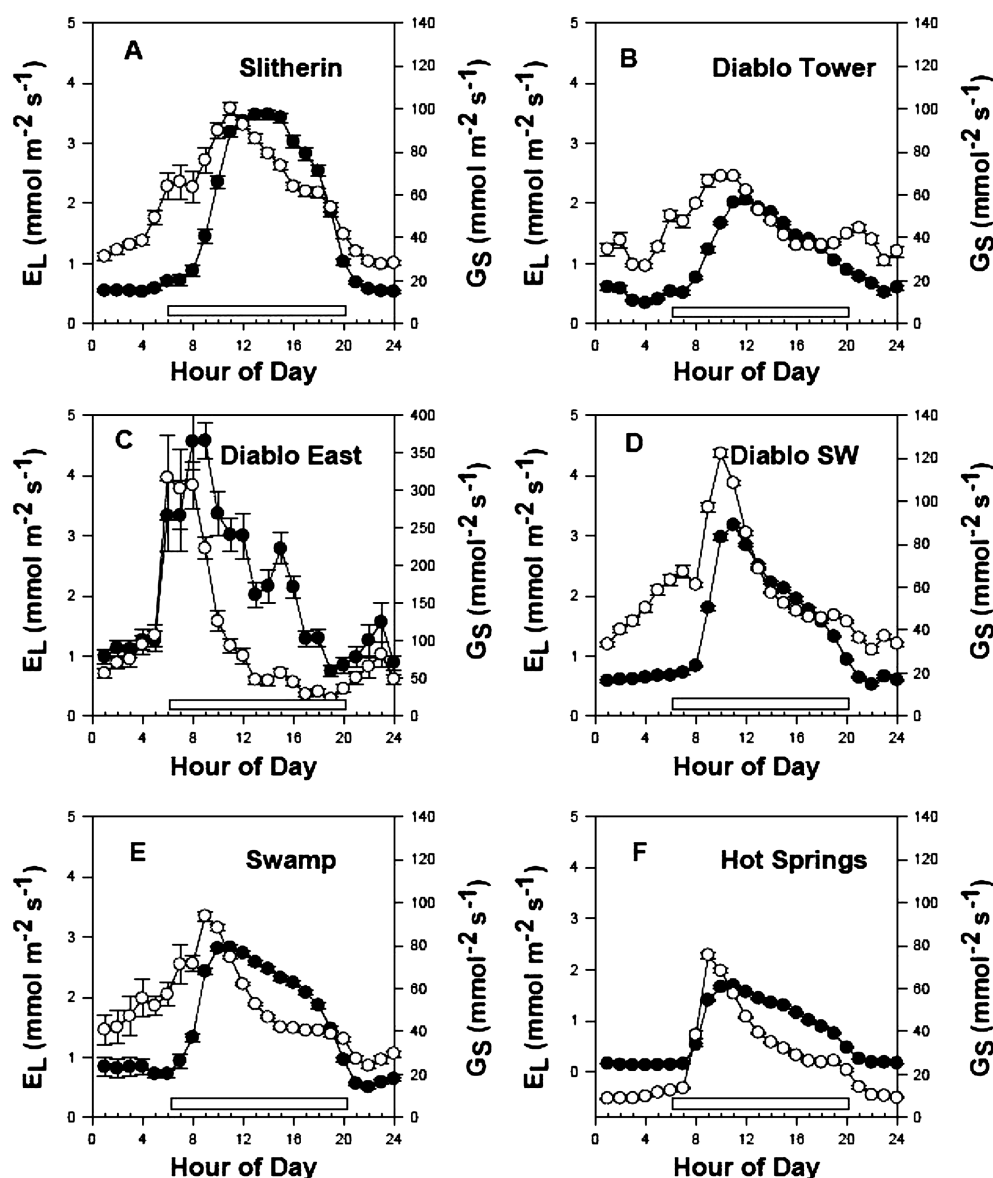


Figure 4. Diurnal traces of leaf-level transpiration (E_L) and stomatal conductance (G_s) for saltcedar measured by sap flux sensors in six separate stands on a river terrace on the Lower Colorado River near Blythe, CA. Error bars are standard errors over 16–35 days of measurement at each site during summer. White bar over the x-axis shows the daylight hours at each site

cloud-free images are prepared for 8-day or 16-day intervals in near real time. Thus, MODIS lends itself to time-series analyses of ET and other physiological processes. Several studies (e.g. Nagler *et al.*, 2005a, 2007; Wang *et al.*, 2007) showed that MODIS EVI was significantly better correlated with ground-based ET measurements than MODIS NDVI across a variety of land cover types, as predicted also by Choudhury *et al.* (1994) in their comparison of NDVI *versus* SAVI. However, non-linear regressions of ET and NDVI were not tested in these studies.

Second, most of them use energy balance flux tower data from the Fluxnet system of towers to calibrate or validate their algorithms (Baldocchi *et al.*, 2001). Ground measurements of ET from flux towers set in natural ecosystems are used to develop best-fit equations between ET, satellite-derived VIs, ancillary remote sensing data and ground meteorological data. However, they vary in

approach from strictly empirical approaches (e.g. Nagler *et al.*, 2009b), to physical models that attempt to determine the parameters in the Penman–Monteith equation with a combination of ground and remote sensing data. For the studies in Table I, r^2 values ranged from 0.45 to 0.95. RMSE of ET ranged from 10% to 25% of mean ET values over the studies.

Examples of K_{c-vi} approaches for mixed biome types

Australia has a national water-monitoring program, developed in part in response to the recent 10-year drought (Glenn *et al.*, this volume). As part of this effort, they reviewed eight operational methods for estimating ET over the continent at a 1 km² scale of measurement. The remote sensing method selected was based on the one developed by Guerschman *et al.* (2009). It used bidirectional-distribution-function corrected MODIS

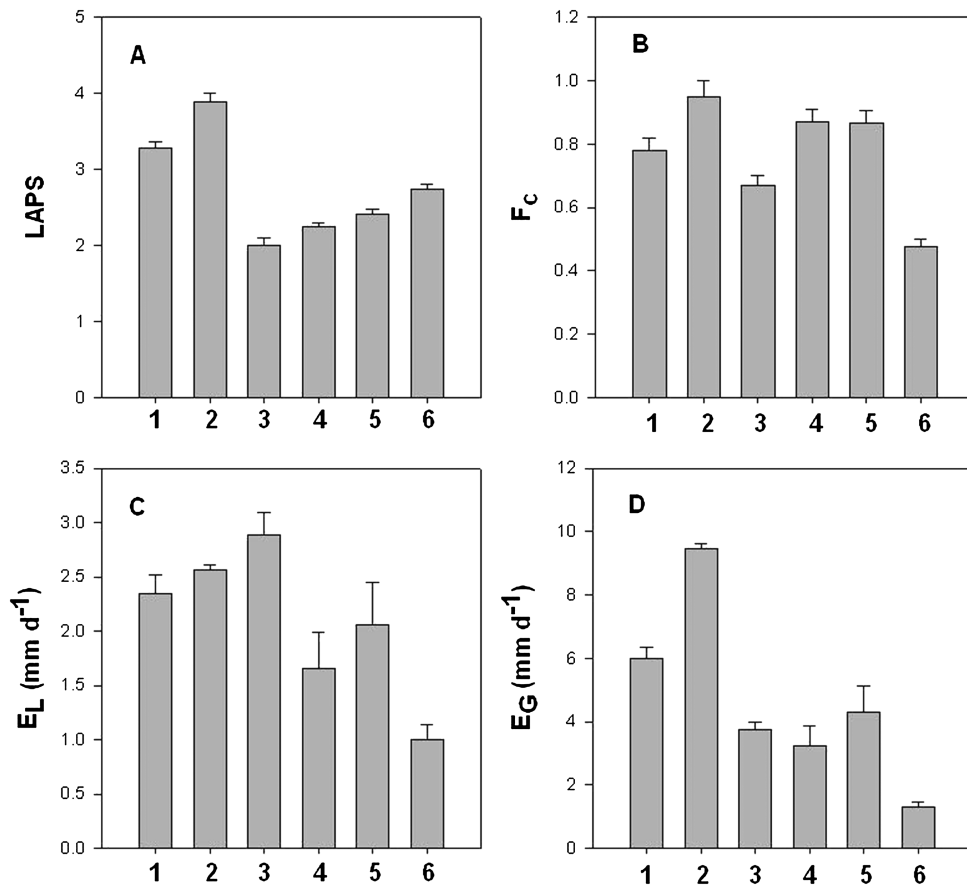


Figure 5. Mean values of plant-specific leaf area index (LAPS) (A); fractional plant cover (F_c) (B); leaf-level transpiration (E_L) (C); and ground-level transpiration (E_G) (D) for saltcedar at six sites on a river terrace on the Lower Colorado River near Blythe, CA. Error bars are standard errors of means. Sites names are Swamp (1), Slitherin (2), Diablo East (3), Diablo Tower (4), Diablo Southwest (5) and Hot Springs (6). LAPS is LAI measured under plant canopies (not including space between canopies). The Hot Springs site had geothermal water intruding into the root zone

products to obtain reflectance values in each band that were independent of view angle, then calculated monthly values of the MODIS-derived Global Vegetation Moisture Index (GVMI) and EVI. They combined the remote sensing data with the Priestly–Taylor model for ET_o (Fisher *et al.*, 2011) to determine ET over Australia. The GVMI estimates surface water content (in plants and soil) by the ratio of reflectance in the NIR to the shortwave IR (SWIR, 1640 nm), based on the strong reflectance in the NIR compared to the SWIR. EVI was used as a measure of LAI, and as an initial assumption, maximum plant transpiration was assumed to be equal to ET_{o-PT} at EVI_{max} and to equal zero at EVI_{min} , as in other methods. ET was scaled between zero and ET_{o-PT} using a sigmoidal rather than a linear equation. An innovation of this method was the use of a Residual Moisture Index to estimate to delineate open water areas and wet soil. EVI and GVMI are expected to be highly correlated if surface moisture is due to vegetation, but they are uncorrelated if surface water is due to wet soil or open water. By plotting the residuals of the EVI:GVMI regressions, vegetated surfaces could be distinguished from unvegetated wet surfaces, and the evaporation from these areas could be calculated separately. This model also incorporated a term for direct evaporation of water from canopies following rain events, based on precipitation data. The model was

then calibrated with ET and other data from seven sites representing grasslands, forests, a lake and a wetland ecosystem. The final model had an r^2 of 0.84 and an RMSE of 23% of the mean measured at flux tower sites, and was able to project ET accurately over the entire content, as compared to water catchment observations.

A NASA group has developed and refined an ET model based on MODIS inputs, gridded meteorological data, and the Penman–Monteith model for ET. The original model of Cleugh *et al.* (2007) was improved by Mu *et al.* (2009) and Mu *et al.* (2011), and is now available as a MODIS Land DAAC product. The method calculated f_c with the MODIS FPAR product. It then calculates ET based on ET_o , f_c and biome-specific values for G_s , aerodynamic resistance and boundary layer resistance derived from a look-up table. The model also accounts for heat losses to the soil, nighttime cuticular transpiration, bare soil evaporation, and direct evaporation of water intercepted by foliage. The algorithm was tested against ET data from 46 moisture flux towers and produced values within 25% of tower values, which is within the range of uncertainty inherent in tower estimates (e.g. Glenn *et al.*, 2007). Note that instead of using ET_o values for a reference crop, this model actually supplies biome-specific values to solve the Penman–Monteith equation.

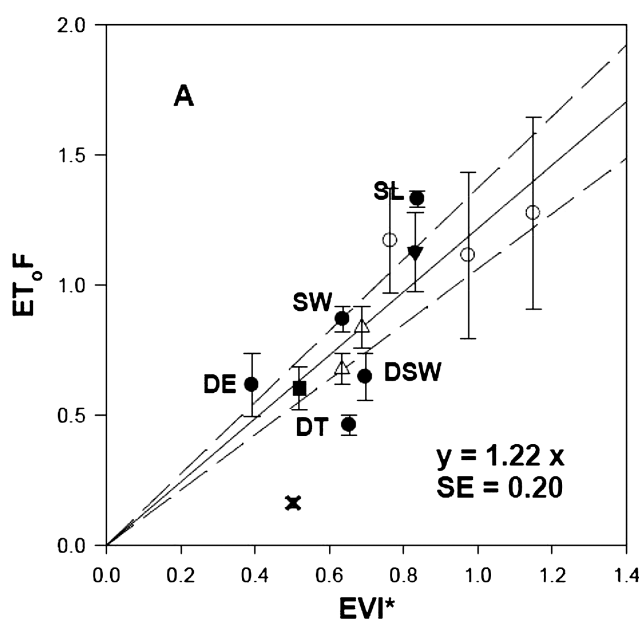


Figure 6. Regression of the ratio of ET to ET_o (ET_o/F) versus the Enhanced Vegetation Index scaled between bare soil (0) and full cover riparian vegetation (1.0) for agricultural and riparian plants in the Lower Colorado River irrigation districts. Solid circles are saltcedar stands at Slitherin, Swamp, Diablo East, Diablo Southwest and Diablo Tower sites near Blythe, CA, measured by sap flow sensors. Open circles are alfalfa measured in a field at Blythe by neutron hydroprobe water balance. Closed square is arrowweed measured by Bowen ratio flux tower at Havasu National Wildlife Refuge (HNWR). Open triangles are saltcedar measured by Bowen ratio towers at HNWR, and the closed triangle is cottonwood measured near Blythe, CA. The X is saltcedar measured at the Hot Springs site near Blythe, excluded from the regression analysis. Error bars are standard errors of means (from Nagler *et al.*, 2009b)

Jung *et al.* (2010) developed a more empirical approach to assess the temporal and spatial behavior of ET for the terrestrial biosphere from 1982–2009, using ET measurements from 253 globally distributed Fluxnet towers, which produced an ET record at point locations from 1997 to 2006. They then combined meteorological data and NDVI from satellite imagery starting in 1982 to created monthly ET maps gridded at 0.5° resolution. The algorithm was developed using a statistical modeling (machine learning) method, and the results were tested against a subset of flux tower data used for validation. They found that global ET generally increased from 1982 to 1998, but has been in decline thereafter, due to long-term droughts in Africa and Australia.

LIMITATIONS OF K_{C-VI} METHODS

The accuracy of VI-derived estimates of ET increase with the accuracy of the ground methods for measuring ET by which they are calibrated. Thus, discrepancies between measured and modeled ET for crops grown in lysimeters were 5%–10%, whereas estimates validated or calibrated by flux tower data were on the order of 10%–30%. This suggests that the error or uncertainty associated with remote sensing estimates of ET is constrained by the accuracy of the ground measurements, and issues with energy balance closure with eddy covariance systems, and if these can be improved, remote sensing estimates will improve as well.

Table I. Examples of vegetation index methods for ET and coefficients of determination with energy balance flux tower data. Entries are arranged by spatial scale of coverage, from local to global

Location	Plant Communities	Explanatory Variables	r^2	Reference
Middle Rio Grande, USA	Arid-zone riparian trees	MODIS EVI, T_a	0.82	Nagler <i>et al.</i> (2005a)
Walnut Gulch, AZ, USA	Desert shrubs and grasses	MODIS EVI, precipitation	0.74	Nagler <i>et al.</i> (2007)
San Pedro River, USA	Riparian trees, shrubs, grasses	MODIS EVI, LST	0.93	Nagler <i>et al.</i> 2005b
Lower Colorado River, USA	Riparian shrubs and trees	MODIS EVI, ET_o	0.80	Nagler <i>et al.</i> (2009a, 2009b)
Three Western USA rivers	Riparian trees, shrubs, grasses	MODIS EVI, T_a	0.76	Nagler <i>et al.</i> (2005b)
Southwestern USA	Phreatophyte shrub communities	Landsat NDVI, ET_o	0.96	Groeneveld <i>et al.</i> (2007)
Southern Great Plains, USA	Native prairie, rangeland, pastures	MODIS EVI, R_n , soil moisture	0.82	Wang and Liang (2008)
Amazonia, Brazil	Rain forest trees	MODIS EVI, R_n	0.72–0.86	Juarez <i>et al.</i> (2008)
Murray-Darling Basin, Australia	Evergreen forests, tropical savannas	MODIS LAI, proxies for variables in Equation (6)	0.66 – 0.90	Zhang <i>et al.</i> (2009a, 2009b)
Pan-Arctic North America	Boreal forests, grasslands, tundra	MODIS LAI, NDVI, ground meteorological data	0.45–0.89	Mu <i>et al.</i> (2009)
Pan-Arctic Basin	Boreal forests, tundra, peatlands	AVHRR NDVI, MODIS Land Cover, solar radiation, albedo	0.91	Zhang <i>et al.</i> (2009a, 2009b)
Australia	Evergreen forests, tropical savannas	MODIS LAI, ground meteorological data	0.74	Cleugh <i>et al.</i> (2007)
Continental-scale North America	Forests, savannas, grasslands, floodplain, lake	MODIS EVI, Global Vegetation Moisture Index, ET_o	0.84	Guerschman <i>et al.</i> (2009)
Continental US	Forests, grasslands, shrublands, croplands	MODIS EVI, LST, solar radiation, land cover class	0.75	Yang <i>et al.</i> (2006)
Global at Fluxnet Sites	Shrublands, grasslands, forests, crops	MODIS EVI, LST, MODIS Land Cover, SWR	0.75	Yang <i>et al.</i> (2006)
Global at Fluxnet Sites	Crops, wetlands, forests, grasslands, savannas	MODIS LAI, proxies for variables in Equation (6)	0.80	Leuning <i>et al.</i> (2008)
Global at Fluxnet Sites	Grasslands, crops, deciduous and evergreen forests	MODIS and AVHRR NDVI, SAVI and EVI, R_n , T_{max} , VPD	0.90	Fischer <i>et al.</i> (2008)
Global at Fluxnet Sites	253 tower sites in all major biomes	AVHRR, MODIS, other satellites, meteorological data	-	Jung <i>et al.</i> (2010)
Global at Fluxnet Sites	46 tower sites in all major biomes	MODIS FPAR, LAI, look-up tables, gridded meteorological data	-	Mu <i>et al.</i> (2011)

The K_{c-VI} methods were most accurate for single agricultural crops, and accuracy can be expected to be less for mixed crops, due to differences in G_s among crops (Allen and Pereira, 2009). Even greater uncertainty is expected for estimates over mixed natural biomes. Nagler *et al.* (2009b) showed that saltcedar, at least, violated some of the major assumptions made in scaling ground estimates of ET by remote sensing. First, most models (e.g. Mu *et al.*, 2011) assume that G_s is zero at night and water loss is only from cuticular transpiration. To the contrary, as much as 26% of total transpiration is through stomata kept open during the night in saltcedar (Moore *et al.*, 2008), apparently as a mechanism for excreting excess salts from leaves (Glenn *et al.*, in press). Second, thermal-band remote sensing methods produce a snapshot of ET at the time of satellite overpass, and they project the results over an entire day or over several days, assuming that $(R_n - G):ET$ is constant over the daylight hours and over different days on a daily time step. However, only one of the six saltcedar sites fulfilled this condition, and the others exhibited midday depression of ET. A mid-morning satellite overpass would overestimate daily ET by different amounts at each site. Hence, an understanding of the environmental constraints on ET for the vegetation of interest in a specific application is needed, along with spatially distributed, seasonal ground measurements of E_L and E_G , to produce a robust ET model using remote sensing. For example, it is still uncertain if ET is more closely related to f_c or LAI, which are predicted differently by VIs.

Despite the limitations, K_{c-VI} methods are increasingly used in tasks such as irrigation scheduling, constructing agricultural district budgets and estimating water use efficiency and opportunities for water conservation in agriculture. They are also used in constructing water budgets for natural or mixed ecosystems at the local, regional and continental scales of measurement, and in projecting trends in global ET over time as affected by climate and land use changes (e.g. Jung *et al.*, 2011). Continuity missions are planned for the major satellite sensor systems used in ET studies, so a seamless trend-line of global ET starting in 1982 can be continued into the future as the hydrological cycle intensifies due to global warming.

REFERENCES

- Albrizio R, Steduto P. 2003. Photosynthesis, respiration and conservative carbon use efficiency of four field grown crops. *Agricultural and Forest Meteorology* **116**: 19–36.
- Alfieri JG, Xiao XM, Niyogi D, Pielke RA, Chen F, LeMone MA. 2009. Satellite-based modeling of transpiration from the grasslands in the Southern Great Plains, USA. *Global Planet Change* **67**: 78–86.
- Allen RG, Pereira LS. 2009. Estimating crop coefficients from fraction of ground cover and height. *Irrigation Science* **28**: 17–34.
- Allen R, Pereira L, Raes D, Smith M. 1998. *Crop evapotranspiration – guidelines for computing crop water requirements* – FAO irrigation and drainage paper 56. Food and Agriculture Organization of the United Nations: Rome, Italy.
- Allen RG, Pereira LS, Smith M, Raes D, Wright JL. 2005. FAO-56 dual crop coefficient method for estimating evaporation from soil and application extensions. *Journal of Irrigation and Drainage Engineering-ASCE* **131**: 2–13.
- Allen R, Tasumi M, Trezza R. 2007. Satellite-based energy balance for mapping evapotranspiration with internalized calibration (METRIC) – Model. *Journal of Irrigation and Drainage Engineering* **133**: 380–394.
- Allen RG, Pereira LS, Howell TA, Jensen ME. 2011. Evapotranspiration information reporting: I. Factors governing measurement accuracy. *Agricultural Water Management* **98**: 899–920.
- Baldocchi D, Falge E, Gu L, Olson R, Hollinger D, Running S, Anthonie P, Berhofer C, Davis K, Evans R, Fuentes J, Goldstein A, Katul G, Law B, Lee X, Malhi Y, Meyers T, Munger W, Oechel W, Pilegaard K, Schmid H, Valentini R, Verma S, Vesala T, Wilson K, Wofsy S. 2001. Fluxnet: a new tool to study the temporal and spatial variability of ecosystem-scale carbon dioxide, water vapor, and energy flux densities. *Bulletin of the American Meteorological Society* **82**: 2415–2434.
- Bannari A, Morin D, Bonn F, Huete A. 1995. A review of vegetation indices. *International Journal of Remote Sensing* **13**: 85–120.
- Barz D, Watson RP, Kanneh JF, Roberts JD, Groeneveld DP. 2009. Cost/benefit considerations for recent saltcedar control, Middle Pecos River, New Mexico. *Environmental Management* **43**: 282–298.
- Bausch WC. 1993. Soil background effects on reflectance-based crop coefficients for corn. *Remote Sensing of Environment* **46**: 213–222.
- Bausch WC. 1995. Remote sensing of crop coefficients for improving the irrigation scheduling of corn. *Agricultural Water Management* **27**: 55–68.
- Bausch WC, Neale CMU. 1987. Crop coefficients derived from reflected canopy radiation: a concept. *Transactions of ASAE* **30**: 703–709.
- Bausch WC, Neale CMU. 1989. Spectral inputs improve corn crop coefficients and irrigation scheduling. *Transactions of ASAE* **32**: 1901–1908.
- Boegh E, Poulsen RN, Butts M, Abrahamsen P, Dellwik E, Hansen S, Hasager CB, Ibrom A, Loeurup JK, Pilegaard K, Soegaard H. 2009. Remote sensing based evapotranspiration and runoff modeling of agricultural, forest and urban flux sites in Denmark: From field to macro-scale. *Journal of Hydrology* **377**: 300–316.
- Campos I, Neale CMU, Calera A. 2011. Assessing satellite-based basal crop coefficients for irrigated grapes (*Vitis vinifera* L.) *Agricultural Water Management* **98**: 45–54.
- Choudhury B, Ahmed N, Idso S, Reginato R, Daughtry C. 1994. Relations between evaporation coefficients and vegetation indices studied by model simulations. *Remote Sensing of Environment* **50**: 1–17.
- Cleugh HA, Leuning R, Mu QZ, Running SW. 2007. Regional evaporation estimates from flux tower and MODIS satellite data. *Remote Sensing of Environment* **106**: 285–304.
- Courault D, Sequin B, Olisio A. 2005. Review on estimation of evapotranspiration from remote sensing data: From empirical to numerical modeling approaches. *Irrigation and Drainage Systems* **19**: 223–249.
- Diak G, Mecikalski J, Anderson M, Norman J, Kustas W, Torn R, DeWolf R. 2004. Estimating land surface energy budgets from space – review and current efforts at the University of Madison – Wisconsin and USDA – ARS. *Bulletin of the American Meteorological Society* **85**: 65–78.
- Duchemin B, Hadria R, Erraki S, Boulet G, Maisongrande P, Chehbouni A, Escadafal R, Ezzahar J, Hoedies JCB, Kharrou MH, Khabba S, Mougenot B, Olisio A, Rodriguez JC, Simoneaux V. 2006. Monitoring wheat phenology and irrigation in Central Morocco: On the use of relationships between evapotranspiration, crop coefficients, leaf area index and remotely-sensed vegetation indices. *Agricultural Water Management* **79**: 1–27.
- El-Shikha DM, Waller P, Hunsaker D, Clarke T, Barnes E. 2007. Ground-based remote sensing for assessing water and nitrogen status of broccoli. *Agricultural Water Management* **92**: 182–193.
- Er-Raki S, Chehbouni A, Guemouria N, Duchemin B, Ezzahar J, Hadria R. 2007. Combining FAO-56 model and ground-based remote sensing to estimate water consumptions of wheat crops in a semi-arid region. *Agricultural Water Management* **87**: 41–54.
- Field C. 1991. Ecological scaling of carbon gain to stress and resource availability. In *Response of plants to multiple stresses*. Mooney H, Winner W, Pell E (eds). Academic Press: London; 35–66.
- Field C, Randerson J, Malmstrom C. 1995. Global net primary production: combining ecology and remote sensing. *Remote Sensing of Environment* **51**: 74–88.
- Fisher JB, Tu KP, Baldocchi DB. 2008. Global estimates of the land-atmosphere water flux based on monthly AVHRR and ISLSCP-II data, validated at 16 FLUXNET sites. *Remote Sensing of Environment* **112**: 901–919.
- Fisher JB, Whittaker RJ, Malhi Y. 2011. ET come home: potential evapotranspiration in geographical ecology. *Global Ecology and Biogeography* **20**: 1–18.
- Glenn EP, Huete AR, Nagler PL, Hirschboeck KK, Brown P. 2007. Integrating remote sensing and ground methods to estimate evapotranspiration. *Critical Reviews in Plant Sciences* **26**: 139–168.

- Glenn EP, Morino K, Didan K, Jordan F, Carroll KC, Nagler PL, Hultine K, Shearer L, Waugh J. 2008. Scaling sap flux measurements of grazed and ungrazed shrub communities with fine and coarse-resolution remote sensing. *Ecohydrology* **1**: 316–329.
- Glenn EP, Nagler PL, Huete AR. 2010. Vegetation index methods for estimating evapotranspiration by remote sensing. *Surveys in Geophysics* **31**: 531–555.
- Glenn EP, Morino K, Nagler PL, Murray S, Pearlstein S, Hultine K. 2012. Role of saltcedar (*Tamarix* spp.) and capillary rise in salinizing a non-flooding terrace on a flow-regulated desert river. *Journal of Arid Environments* (in press).
- Gontia NK, Tiwari KN. 2010. Estimation of crop coefficient and evapotranspiration of wheat (*Triticum aestivum*) in a irrigation command using remote sensing and GIS. *Water Resources Management* **24**: 1399–1414.
- Gonzalez-Dugo MP, Mateos L. 2008. Spectral vegetation indices for benchmarking water productivity of irrigated cotton and sugarbeet crops. *Agricultural Water Management* **95**: 48–58.
- Gonzalez-Dugo MP, Neale CMU, Mateos L, Kustas WP, Prueger JH, Anderson MC, Li F. 2009. A comparison of operational remote sensing-based models for estimating crop evapotranspiration. *Agricultural and Forest Meteorology* **149**: 1843–1853.
- Gowda PH, Chavez JL, Colaizzi PD, Evett SR, Howell TA, Tolk JA. 2008. ET mapping for agricultural water management: present status and challenges. *Irrigation Science* **26**: 223–237.
- Groeneveld DP, Baugh WM. 2007. Correcting satellite data to detect vegetation signal for eco-hydrologic analyses. *Journal of Hydrology* **344**: 135–145.
- Groeneveld DP, Baugh WM, Sanderson JS, Cooper DJ. 2007. Annual groundwater evapotranspiration mapped from single satellite scenes. *Journal of Hydrology* **344**: 146–156.
- Guerschman JP, Van Dijk AIJM, Mattersdorf G, Beringer J, Hutley LB, Leuning R, Pipunic RC, Sherman BS. 2009. Scaling of potential evapotranspiration with MODIS data reproduces flux observations and catchment water balance observations across Australia. *Journal of Hydrology* **369**: 107–119.
- Huete AR. 1988. A soil adjusted vegetation index (SAVI). *Remote Sensing of Environment* **25**: 295–309.
- Huete AR, Glenn EP. 2011. Remote sensing of ecosystem structure and function. In *Advances in Environmental Remote Sensing*, Weng Q (ed). CRC Press: Boca Raton, Florida; 291–320.
- Huete A, Didan K, Miura T, Rodriguez E, Gao X, Ferreira L. 2002. Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment* **83**: 195–213.
- Hunsaker DJ, Pinter PJ, Barnes EM, Kimball BA. 2003. Estimating cotton evapotranspiration crop coefficients with a multispectral vegetation index. *Irrigation Science* **22**: 95–104.
- Hunsaker DJ, Pinter PJ, Kimball BA. 2005a. Wheat basal crop coefficients determined by normalized difference vegetation index. *Irrigation Science* **24**: 1–14.
- Hunsaker DJ, Barnes EM, Clarke TR, Fitzgerald GJ, Pinter PJ. 2005b. Cotton irrigation scheduling using remotely sensed and FAO-56 basal crop coefficients. *Transactions of ASAE* **48**: 1395–1407.
- Hunsaker DJ, Fitzgerald GJ, French AN, Clarke TR, Ottman MJ, Pinter PJ. 2007a. Wheat irrigation management using multispectral crop coefficients. I. Crop evapotranspiration prediction. *Transactions of the ASABE* **50**: 2017–2033.
- Hunsaker DJ, Fitzgerald GJ, French AN, Clarke TR, Ottman MJ, Pinter PJ. 2007b. Wheat irrigation management using multispectral crop coefficients. II. Irrigation scheduling performance, grain yield, and water use efficiency. *Transactions of the ASABE* **50**: 2035–2050.
- Hunsaker DJ, El-Shikha DM, Clarke TR, French AN, Thorp KR. 2009. Using ESAP software for predicting the spatial distributions of NDVI and transpiration of cotton. *Agricultural Water Management* **96**: 1293–1304.
- Hunsaker DJ, French AN, Clarke TR, El-Shikha DM. 2011. Water use, crop coefficients and irrigation management criteria for camelina production in arid regions. *Irrigation Science* **29**: 27–43.
- Huxman T, Smith M, Fay P, Knapp S, Shaw M, Loik M, Smith D, Tissue C, Zak J, Weltzin J, Pockman W, Sala O, Haddad B, Harte J, Koch G, Schwinning S, Small E, Williams D. 2004. Convergence across biomes to a common rain-use efficiency. *Nature* **429**: 651–654.
- Jayanathi H. 2003. Airborne and ground-based remote sensing for the estimation of evapotranspiration and yield of bean, potato and sugar beet. PhD dissertation, Utah State University, Logan, Utah.
- Jayanathi H, Neale CMU, Wright JL. 2000. Seasonal evapotranspiration estimation using canopy reflectance - A case study. *IAHS Proceedings of Remote Sensing and Hydrology, 2000*, April 3–5, Sante Fe, New Mexico.
- Jayanathi H, Neale CMU, Wright JL. 2007. Development and validation of canopy reflectance-based crop coefficients for potato. *Agricultural Water Management* **88**: 235–246.
- Jensen ME. 1968. Water consumption by agricultural plants. In *Water deficits and plant growth*, vol. 2, Chapter 1, Kozlowski TT (ed). Academic Press: Salt Lake City, Utah, 1–22 pp.
- Jensen M. 1998. *Coefficients for vegetative evapotranspiration and open water evaporation for the Lower Colorado River Accounting System*. United States Bureau of Reclamation, Boulder Canyon Operations Office: Boulder City, Nevada.
- Jensen ME, Wright JL, Pratt BJ. 1971. Estimating soil moisture depletion from climate, crop and soils data. *Transactions of ASAE* **14**: 954–959.
- Johnson L. 2010. Information technology supports integration of satellite imagery with irrigation management. *5th National Decennial Irrigation Conference*, Paper No. IRR10-9645, ASABE Publication 711P081cd.
- Juarez RIN, Goulden ML, Myneni RB, Fu R, Bernardes S, Gao H. 2008. Estimating catchment evapotranspiration and runoff using MODIS leaf area index and the Penman-Monteith equation. *International Journal of Remote Sensing* **29**: 7045–7063.
- Jung M, Reichstein M, Ciais P, Seneviratne SI, Sheffield J, Goulden ML, Bonon G, Cescatti A, Chen JQ, de Jeu R, Dolman AJ, Eugster W, Gerten D, Gianelle D, Gobron N, Heinke J, Kimball J, Law BE, Montagnini L, Mu QZ, Mueller B, Oleson K, Papale D, Richardson AD, Rouspard O, Running S, Tomelleri E, Viovy N, Weber U, Williams C, Wood E, Zaehle S, Shang K. 2010. Recent decline in the global land evapotranspiration trend due to limited moisture supply. *Nature* **467**: 951–954.
- Kalma JD, McVicar TR, McCabe MF. 2008. Estimating land surface evaporation: A review of methods using remotely sensed surface temperature data. *Surveys in Geophysics* **29**: 421–469.
- Kustas W, Anderson M. 2009. Advances in thermal infrared remote sensing for land surface modeling. *Agricultural and Forest Meteorology* **149**: 2071–2081.
- Leuning R, Zhang YQ, Rajaud A, Cleugh H, Tu K. 2008. A simple surface conductance model to estimate regional evaporation using MODIS leaf area index and the Penman-Monteith equation. *Water Resources Research* **44**: W10419.
- Li R, Min QL, Lin B. 2009. Estimation of evapotranspiration in a mid-latitude forest using the Microwave Emissivity Difference Vegetation Index (EDVI). *Remote Sensing of Environment* **113**: 2011–2018.
- Mata-Gonzalez R, McLendon T, Martin DW. 2005. The inappropriate use of crop transpiration coefficients (K_c) to estimate evapotranspiration in arid ecosystems. *Arid Land Research and Management* **19**: 285–295.
- Moore GW, Cleverly JR, Owens MK. 2008. Nocturnal transpiration in riparian Tamarix thickets authenticated by sap flux, eddy covariance and leaf gas exchange measurements. *Tree Physiology* **28**: 521–528.
- Mu QZ, Zhao M, Running SW. 2011. Improvements to a MODIS global terrestrial evapotranspiration algorithm. *Remote Sensing of Environment* (in press). Available on-line as DOI: 10.1016/res.2011.02.019
- Murray RS, Nagler PL, Morino K, Glenn EP. 2009. An empirical algorithm for estimating agricultural and riparian evapotranspiration using MODIS Enhanced Vegetation Index and ground measurements of ET. II. Application to the Lower Colorado River, U.S. *Remote Sensing* **1**: 1125–1138.
- Myneni RB, Hoffman S, Knyazikhim Y, Privette JL, Glassy J, Tian Y, Wang Y, Song X, Zhang Y, Smith GR, Lotsch A, Friedl M, Morisette JT, Votava P, Nemani RR, Running SW. 2002. Global products of vegetation leaf area index and fraction absorbed PAR from year one of MODIS data. *Remote Sensing of Environment* **83**: 214–231.
- Nagler P, Glenn E, Huete A. 2001. Assessment of vegetation indices for riparian vegetation in the Colorado River delta, Mexico. *Journal of Arid Environments* **49**: 91–110.
- Nagler P, Cleverly J, Lampkin D, Glenn E, Huete A, Wan, Z. 2005a. Predicting riparian evapotranspiration from MODIS vegetation indices and meteorological data. *Rem Sens Envir* **94**: 17–30.
- Nagler P, Scott R, Westenberg C, Cleverly J, Glenn E, Huete A. 2005b. Evapotranspiration on western U.S. rivers estimated using the Enhanced Vegetation Index from MODIS and data from eddy covariance and Bowen ratio flux towers. *Remote Sensing of Environment* **97**: 337–351.
- Nagler P, Glenn E, Kim H, Emmerich W, Scott R, Huxman T, Huete, A. 2007. Seasonal and interannual variation of ET for a semiarid watershed estimated by moisture flux towers and MODIS vegetation indices. *Journal of Arid Environments* **70**: 443–463.
- Nagler PL, Morino K, Didan K, Osterberg J, Hultine K, Glenn EP. 2009a. Wide-area estimates of saltcedar (*Tamarix* spp.) evapotranspiration on the lower Colorado River measured by heat balance and remote sensing methods. *Ecohydrology* **2**: 18–33.

- Nagler PL, Morino K, Murray R, Osterberg J, Glenn EP. 2009b. An empirical algorithm for estimating agricultural and riparian evapotranspiration using MODIS Enhanced Vegetation Index and ground ground measurements of ET. I. Description of method. *Remote Sensing* **1**: 1273–1297.
- Neale CMU. 1983. Monitoring corn development using reflected radiation. M.S. Thesis, Colorado State University, Fort Collins, 134 pp.
- Neale CMU. 1987. Development of Reflectance-based crop coefficients for Corn. PhD Dissertation, Colorado State University, Fort Collins.
- Neale CMU, Bausch WC, Heermann DF. 1989. Development of reflectance-based crop coefficients for corn. *Transactions of ASAE* **32**: 1891–1899.
- Neale CMU, Ahmed RH, Moran MS, Pinter, Jr. PJ, Qi J, Clarke TR. 1996. Estimating cotton seasonal evapotranspiration using canopy reflectance. *Proceedings of the International Conference on Evapotranspiration and Irrigation Scheduling*, 3–6 November, San Antonio, Texas.
- Neale CMU, Hipps HP, Prueger JH, Kustas WP, Cooper D, Eichinger WE. 2001. Spatial mapping of evapotranspiration and energy balance components over riparian vegetation using airborne remote sensing. *Remote Sensing and Hydrology 2000*, IAHS Publication no. 267.
- Neale C, Jayanthi H, Wright JL. 2005. Irrigation water management using high resolution airborne remote sensing. *Irrigation and Drainage Systems* **19**: 321–336.
- Rafn EB, Contor B, Ames DP. 2008. Evaluation of a method for estimating irrigated crop-evapotranspiration coefficients from remotely sensed data in Idaho. *Journal of Irrigation and Drainage Engineering ASCE* **134**: 722–729.
- Rouse JW, Hass RH, Schell JA, Deering DW. 1973. Monitoring vegetation systems in the Great Plains with ERTS. *Third ERTS Symposium, NASA SP-351* **1**: 309–317.
- Samani Z, Bawazir AS, Bleiweiss M, Skaggs R, Longworth J, Tran VD, Pinon A. 2009. Using remote sensing to evaluate the spatial variability of evapotranspiration and crop coefficient in the lower Rio Grande Valley, New Mexico. *Irrigation Science* **28**: 93–100.
- Santos C, Lorite IJ, Tasumi M, Allen RG, Fereres E. 2007. Integrating satellite-based evapotranspiration with simulation models for irrigation management at the scheme level. *Irrigation Science* **3**: 277–288.
- Scheffeld J, Ferguson CR, Troy TJ, Wood EF, McCabe MF. 2009. Closing the terrestrial water budget from satellite remote sensing. *Geophysical Research Letters* **36**: L07403.
- Scott RL, Cable WL, Huxman TE, Nagler PL, Hernandez M, Goodrich DC. 2008. Multiyear riparian evapotranspiration and groundwater use for a semiarid watershed. *Journal of Arid Environments* **72**: 1232–1246.
- Shuttleworth WJ. 2007. Putting the 'vap' into evapotranspiration. *Hydrology and Earth System Sciences* **11**: 210–244.
- Singh RK, Irmak A. 2009. Estimation of crop coefficients using satellite remote sensing. *Journal of Irrigation and Drainage Engineering ASCE* **135**: 597–608.
- Tang QH, Gao HL, Lu H, Lettenmaier DP. 2009. Remote sensing: hydrology. *Progress in Physical Geography* **33**: 490–509.
- Trout T, Johnson L, Gartun J. 2008. Remote sensing of canopy cover in horticultural crops. *HortScience* **43**: 333–337.
- Walter IA, Allen RG, Elliott R, Itenfisu D, Brown P, Jensen ME, Mecham B, Howell TA, Snyder RL, Echling S, Spofford T, Hattendorf M, Martin D, Cuenca RH, Wright JL. 2002. The ASCE standardized reference evapotranspiration equation. American Society of Civil Engineering, Reston, VA. <http://www.kimberly.uidaho.edu/water/asceewri/main.pdf>
- Wang KC, Liang SL. 2008. An improved method for estimating global evapotranspiration based on satellite determination of surface net radiation, vegetation index, temperature and soil moisture. *Journal of Hydrometeorology* **9**: 712–727.
- Wang KC, Wang P, Li ZQ, Cribb M, Sparrow M. 2007. A simple method to estimate actual evapotranspiration from a combination of net radiation, vegetation index, and temperature. *Journal of Geophysical Research – Atmosphere* **112**: D151107.
- Westenburger CL, Harper DP, DeMeo GA. 2006. Evapotranspiration by phreatophytes along the Lower Colorado River at Havasu National Wildlife Refuge, Arizona. U.S. Geological Survey Scientific Investigations Report 2006–5043 (<http://pubs.usgs.gov/sir/2006/5043/>).
- Williams D, Cable W, Hultine K, Hoedjes H, Yezpe E, Simonneaux V, Er-Raki S, Boulet G, de Bruin H, Chehbouni A, Hartogensis O, Timouk F. 2004. Evapotranspiration components determined by stable isotope, sap flow and eddy covariance techniques. *Agricultural and Forest Meteorology* **125**: 241–258.
- Wright JL. 1982. New evapotranspiration crop coefficients. *Journal of the Irrigation and Drainage Division ASCE* **108**(IR2): 57–74.
- Xu D, Shen Y. 2005. External and internal factors responsible for midday depression of photosynthesis. In *Handbook of photosynthesis*, 2nd edn, Pessarakli M (ed). Taylor & Francis: Boca Raton, FL, USA; 297–298.
- Xu CY, Singh VP. 2002. Cross comparison of empirical equations for calculating potential evapotranspiration with data from Switzerland. *Water Resources Management* **16**: 197–219.
- Zhang F, White M, Michaelis A, Ichii K, Hashimoto H, Votava P, Zhu S, Nemani R. 2006. Prediction of continental-scale evapotranspiration by combining MODIS and Ameriflux data through support vector machine. *IEEE Transactions in Geosciences and Remote Sensing* **44**: 3452–3461.
- Zhang K, Kimball JS, Mu QZ, Jones LA, Goetz SJ, Running SW. 2009a. Satellite based analysis of northern ET trends and associated changes in the regional water balance from 1983 to 2005. *Journal of Hydrology* **379**: 92–110.
- Zhang YQ, Chew FHS, Zhong L, Li HX. 2009b. Use of remotely sensed actual evapotranspiration to improve rainfall-runoff modeling in Southeast Australia. *Journal of Hydrology and Meteorology* **10**: 969–980.